Blockchain for Energy Credits and Certificates: A Comprehensive Review

Syed Muhammad Danish, Kaiwen Zhang, Fatima Amara, Juan Carlos Oviedo Cepeda, Luis Fernando Rueda Vasquez, and Tom Marynowski

***Abstract*—Climate change is a major issue that has disastrous impacts on the environment through different causes like the greenhouse gas (GHG) emission. Many energy utilities around the world intend to reduce GHG emissions by promoting different systems including carbon emission trading (CET), renewable energy certificates (RECs), and tradable white certificates (TWCs). However, these systems are centralized, highly regulated, and operationally expensive and do not meet transparency, trust and security requirements. Accordingly, GHG emission reduction schemes are gradually moving towards blockchain-based solutions due to their underpinning characteristics including decentralization, transparency, anonymity, and trust (independent from third parties). This paper performs a comprehensive investigation into the blockchain technology, deployed for GHG emission reduction plans. It explores existing blockchain solutions along with their associated challenges to effectively uncover their potentials. As a result, this study suggests possible lines of research for future enhancements of blockchain systems particularly their incorporation in GHG emission reduction.**

***Index Terms*—Carbon trading, energy trading, green certificates, white certificates, renewable energy resources.**

I. Introduction

OVER the past few decades, energy has been a vital resource driving our economy. With increasing digitalization and sophistication of our society, our dependence on energy has also grown. Fossil fuels, which make up 80% of the world's energy, remain the dominant energy source with petroleum, coal, and natural gas accounting for a quarter of the world's energy needs [1], [2]. However, fossil fuels are a significant contributor to environmental pollution and traditional centralized energy systems based on fossil fuels face numerous challenges for example energy crises, environmental pollution, long-distance transmission, and carbon emissions [3]. Due to these issues, there has been a greater emphasis on exploring alternative energy sources [4], particularly renewable energy such as solar and wind energy, which are considered clean and environmentally friendly resources that are critical in addressing environmental impact of carbon emissions [5], [6].

Many countries and energy utilities worldwide are currently aiming to reduce greenhouse gas emissions significantly and invest in cleaner options. To achieve this goal, several methods have been proposed, such as CETs, RECs, and TWCs, etc. Carbon trading [7] enables institutions that are unable to reduce their emissions to compensate by purchasing credits from others who have met their emission reduction targets. This allows these institutions to improve their reputation in terms of environmental impact. For example, an oil company that exceeds the emissions limit may need to purchase or use saved credits to stay within the emissions cap. Green energy certificates [8], on the other hand, are provided by companies that generate electricity using renewable energy sources, enabling them to sell green energy and receive subsidies. For financial rebates, homeowners can install renewable energy resources in their homes and generate solar photovoltaic power. Finally, tradeable white certificates [9] are tradable assets that verify that energy savings have been achieved relative to a baseline.

In order to reduce global greenhouse gas emissions, proposed systems must consider transparency, security, safety, and registration of energy data from trading processes, e.g., CETs and RECs. However, there are challenges associated with managing a large number of dynamic participants in such systems, ranging from small-scale energy producers to large companies. Centralized management systems, commonly used in traditional trading systems, are not scalable or suitable for managing a large number of participants [10], and the annual audit cost for energy sources and carbon credits is prohibitively high, which often excludes small-scale energy producers from the market [11]. Additionally, these trading systems are susceptible to fraud due to the underlying complexity of the system, lack of transparency, and mutual trust between participants [12]. Finally, centralized trading systems can compromise participants' privacy, especially for small-scale producers and traders who prefer not to disclose their trade information and market exposure level [11].

We discussed the importance of decentralization, transparency, trust, anonymity, and reliability in carbon trading and energy certificates, the traits commonly associated with the blockchain technology [13]. These traits have been utilized in various applications, including supply chain management, healthcare, transportation, the Internet of Things (IoT), and

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TABLE I
OVERVIEW OF RELATED SURVEYS IN ENERGY TRADING

Work(s)	Major Contributions of the Survey(s)
[18]–[23]	Blockchain's application in smart grid.
[24]	Role of the blockchain-based microgrid in energy communities' operation and trading.
[10], [25]–[28]	Blockchain-based decentralized, wholesale and peer-to-peer (P2P) energy trading mechanisms
[29]	Blockchain-based trading in electric power systems.
[30]	P2P energy trading through blockchain in existing power sector.
[31]	Blockchain applications in sustainable energy systems.
This work	Blockchain application in GHG emission reduction schemes including CET, RECs, TWCs, etc.

energy trading, and [14], [15], [16], [17]. Unlike traditional systems, blockchain technology is decentralized, which does not require a central trusted authority. Multiple blockchain nodes in the network interact with one another to create, store, and maintain a chain of tamper-proof blocks. Every entity in the network can verify the chain order and stored data correction, which lead to transparency and trust.

For carbon trading and energy certifications, blockchain technology is becoming increasingly popular as a means to ensure decentralization. This article seeks to present an overview of the recent blockchain applications in various global greenhouse gas emission reduction mechanisms, such as CETs, RECs, TWCs, and others.

Related Surveys and Our Contributions: Recently, researchers have conducted several works to address energy trading issues by blockchain. Until now, several surveys have been published in [10], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], where the authors attempted to review the blockchain application in energy trading, smart grid, microgrid, and energy management. The authors in [18], [19], [20], [21], [22], [23] provide a comprehensive review of blockchain applications in the smart grid. Authors in [24] discusses the role of microgrid and blockchain technologies in energy community operation and trading considering cybersecurity, energy flexibility, and the cooperative market. Surveys on the use of blockchain technology in the energy sector are presented in [10], [25], [26], [27], [28], which cover decentralized, wholesale, and peer-to-peer (P2P) energy trading aspects using the blockchain network. Authors in [29] investigate the blockchain-based energy trading in the electrical power system. In [30], a detailed analysis is presented on how the current power sector is being transformed by employing blockchain technology towards P2P energy trading. Additionally, [31] provides an overview of the ways in which blockchain technology is being applied in sustainable energy systems. Table I summarizes the main contributions of this article compared to others.

However, the previously conducted studies focused only on specific areas such as smart grids, microgrids, or energy trading, and only on limited topics. Currently, there is no comprehensive survey that covers the latest advancements in blockchain technology applications in energy certifications and carbon trading. This motivated us to provide a comprehensive and in-depth survey of the state-of-the-art research on blockchain applications in energy certifications and carbon trading. The main contributions of this article are listed below:

1) We discuss the major requirements and key research challenges associated with these trading mechanisms in terms of trust, transparency, security, decentralization, and privacy, and to realize how blockchain can contribute to solving those challenges.
2) We also present a comprehensive overview of existing blockchain-based carbon trading and energy certification solutions. Moreover, this work also highlights the state-of-the-art issues and techniques investigated using the blockchain technology.
3) Lastly, we discuss the present research gaps in state-of-the-art, and prospects for future blockchain research in the field of energy certification and carbon trading.

The rest of the paper is organized as follows. First, Section II introduces the GHG emission reduction schemes. Sections III and IV reviews how blockchain is applied in CET and energy certificates, respectively. Then, in Section V, we identify the current challenges and possible future research directions. Finally, we conclude this article in the last section.

II. BACKGROUND

A. Blockchain

The blockchain can be viewed as a public ledger, where all committed transactions are stored in a chain of blocks. With the addition of new blocks, this chain continuously grows. Blockchain technology functions in a decentralised environment based on several core technologies such as cryptographic hashing, digital signatures (based on asymmetric cryptography), and distributed consensus mechanisms. With blockchain technology, a transaction can take place in a decentralised fashion. Essentially, the blockchain is a sequence of blocks that holds a complete list of transaction records. There is a reference in every block pointing to the immediately previous block, called a parent block, which is essentially a hash value of the previous block. Blocks are added to the blockchain after transactions are validated, and once the network confirms the addition of a block, it cannot be modified. There are two general categories of blockchain approaches: permissionless and permissioned. Permissionless blockchain networks allow anyone to read and write the blockchain without authorization. The permissioned blockchain network allows for finer-grained controls and limits participation to specific individuals or organizations. Fig. 1 illustrates an example of blockchain-based emission reduction literature.

B. Carbon Emission Trading

The Paris Agreement, signed in December 2015, brought all countries together to make an effort to fight against climate change by committing to nationally determined contributions [32]. One approach to reduce greenhouse gas emissions is to implement an emission trading system, which creates a carbon market where carbon dioxide emission is treated as a commodity,

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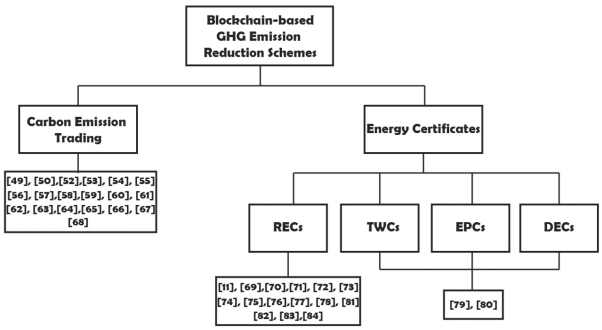


Fig. 1. GHG emission reduction schemes.

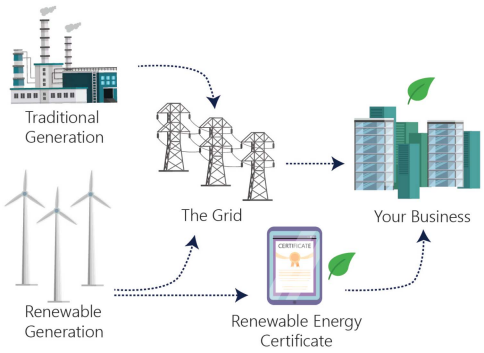


Fig. 2. Renewable energy certificates.

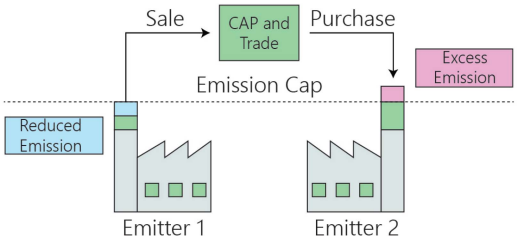


Fig. 3. Carbon emission trading system [36].

resulting in the production of carbon credits [33]. These carbon credits are certificates that can be bought and sold, allowing the holder to release one metric ton of carbon dioxide or an equivalent amount of greenhouse gas emission, as illustrated in Fig. 3. Carbon offsetting enables individuals and businesses to lower their carbon emissions by purchasing carbon credits from projects that lower carbon production, such as reforestation, renewable energy investments, carbon capture, and sequestration projects. As an example, let's say "Company A" is given a limit of 100 units of greenhouse gas (GHG) emissions by a regulatory body but emits 120 units by the end of the year. "Landowner B" has conserved a forest, entitling them to 50 carbon offset credits. "Company A" can purchase 20 offset credits from "Landowner B" at the market rate to increase their emission cap.

Development Context: The EU ETS operates through distinct "trading periods." The initial ETS trading period spanned three years, commencing in January 2005 and concluding in December 2007. Subsequently, the second trading period encompassed the period from January 2008 to December 2012, coinciding with the first commitment phase of the Kyoto Protocol. Following this, the third trading period extended from January 2013 to December 2020. In comparison to 2005, the proposed emission caps for 2020 represented a notable 21% reduction in greenhouse gases. The fourth phase started in January 2021 and will continue until December 2030. Research efforts have also been directed towards addressing VAT fraud and cybercrimes that have targeted the ETS, focusing on enhancing security and transparency within the system [11].

C. Energy Certificates

An energy certificate is a certification that can be transferred and can come in any form, including electronic, indicating the amount of a particular type of energy or material goods that are consumed or saved by a device or building that converts energy. The various kinds of energy certificates are detailed below.

1) Renewable Energy Certificates: A REC is an electronic certificate granted to the renewable energy generators by the government, who fulfill certain conditions, as depicted in Fig. 2. The certificate is a means of acknowledging power generation companies for producing renewable energy and serves as proof for the buyers that they are consuming green electricity. As electricity generated from renewable sources is not distinguishable from electricity produced from other sources, some form of monitoring is required. These certificates provide proof that the owner of an energy market instrument possesses one megawatt hour (MWh) of renewable energy, representing the amount of renewable energy that passes through the power grid. These certificates can be sold for profit to individuals wishing to offset their carbon emissions or to speculators who wager on the value of energy credits.

2) Tradable White Certificates: A white certificate [9], also known as Energy Savings Certificate (ESC) or an Energy Efficiency Credit (EEC), is a type of tradable asset issued by the government that serves as proof that a certain percentage of energy savings has been achieved relative to a baseline. Retail energy distributors or suppliers are required to achieve a specific amount of energy savings, and by saving the energy in different projects beyond what is typically expected, they can create tradeable white certificates. The market actor can sell them to other obligated parties,or use the certificates for their own target compliance. It's important to note that a white-certificate program may not involve buying and selling, and can be thought of as similar to a green-certificate program that is awarded for reaching a minimum amount of renewable energy production.

3) Energy Performance and Display Energy Certificates: The Energy Performance Certificate (EPC) [34] is a document that evaluates the energy efficiency of a property using a rating scale from A-G, where A indicates the most efficient homes with lower carbon dioxide emissions. An EPC is legally required for a property to be sold, let, or constructed, and it remains valid for ten years. It indicates the amount of CO2 emissions produced by the property in tons and suggests ways to reduce the emission amount.

On the other hand, Display Energy Certificates (DECs) [35] are intended to enhance transparency about the energy efficiency of public buildings. They were introduced by several European

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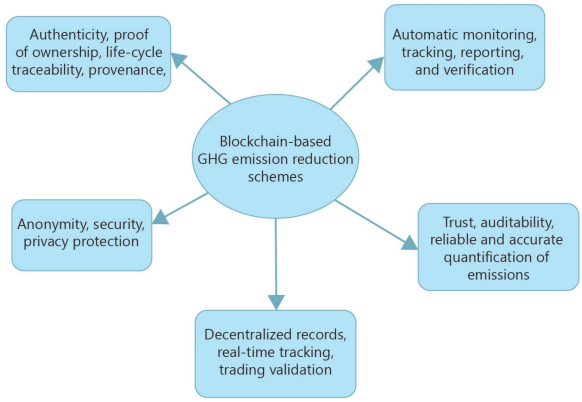


Fig. 4. Benefits of a blockchain-based CET system.

governments and record the actual energy consumption of public buildings, using the same A-G scale as EPCs. DECs measure the energy usage of a building based on its annual consumption, while EPCs assess the energy efficiency of a building based on factors such as insulation, heating, lighting, and ventilation.

Transparency in GHG Emission Reduction Systems: In the context of climate change or greenhouse gas emissions, transparency refers to the reliable measurement and trusted reporting of data for accountability among nations, organizations, and industries. When emissions data is openly shared and verified, it builds trust among stakeholders and demonstrates a commitment to addressing climate change. For example, when emissions are accurately reported, each participant receives their allocated allowances based on their actual emissions, preventing overallocation or under-allocation of allowances that could distort market dynamics. In the context of GHG emission reduction trading, the transparent reporting of transactions between buyers and sellers ensures the integrity of the market by preventing fraud, misreporting, and manipulation of emissions data. This maintains trust among participants and prevents the system from being compromised. For example, without transparency, participants might manipulate emissions data to their advantage, may double spend their emission certificates or can engage in market manipulation by hiding their trading activities, etc. Therefore, transparency in GHG emission reduction schemes is essential to maintain the overall credibility and fairness of the monitoring, reporting, verification, and trading mechanisms in the system.

III. Blockchain Contributions in Energy Credits

This section reviews how blockchain is applied in carbon trading by presenting several recent works. Fig. 4 provides a brief overview of benefits of blockchain in GHG emission reduction schemes. Moreover, a summary of how blockchain has been utilized for energy credits is provided in Table II.

A. Blockchain for Carbon Emission Trading

Currently, the CET market faces several challenges. First, carbon budget calculations are at risk of being manipulated and abused, while the centralized approval process results in high administration and transaction costs [37]. Second, the carbon trading market lacks transparency, with companies operating without oversight, and their systems and decisions not being accessible. To address this, companies should be required to provide transparency and record, report, and verify emission transactions, ensuring their transparency and unforgeability [38]. This would improve the efficiency of the carbon market by reducing uncertainty. Furthermore, because carbon emission permits are traded so frequently, tracing becomes extremely difficult, so it's imperative to keep tamper-proof records of carbon emission quotas, which could be provided by blockchain-based solutions. As centralized architectures require central institutions or central technical systems, the lack of privacy protection and information security for traders is one of the major obstacles to the widespread adoption of carbon trading systems. In light of these challenges, developing a traceable, secure, and trusted decentralized carbon trading system is crucial, and blockchain studies offer a potential solution. We now summarize some related blockchain studies to show how blockchain can be employed to solve the above mentioned issues.

Recent works have emphasized how important the blockchain technology is, to solve the above mentioned issues. Authors in [37], [39], [40], [41], [42], [43] argued that the distinct characteristics of blockchain and smart contracts can enable automatic monitoring, tracking, reporting, and verification of carbon emissions. Moreover, blockchain-based solution can also help validate the authenticity, provenance, transparency, and reliability of the overall carbon trading system along with the protection of the privacy of traders [44], [45], [46], [47], [48]. In [49], the authors suggested a carbon trading system that incorporates reputation to improve efficiency and prevent fraud. The reputation or merit score would be tracked on a distributed ledger to monitor the success of carbon reduction programs in participating industries. In [50], the authors proposed a hybrid emissions trading scheme using smart contracts on a permissioned blockchain to ensure transparency, tamper-resistance, and high liquidity. This system is designed to be both backward compatible and future-proof. Smart contracts are used to implement the carbon trading mechanism with the help of Bancor AMM [51] protocol.

Recently, authors in [52] argued that the carbon credit markets are expanding from from B2B (business-to-business) markets to P2P markets, and proposed a delegated proof-of-stake-based carbon emission rights verification system using blockchain technology to learn the data and to verify these carbon emissions of individual entities with the help of anomaly detection technique. The fashion apparel manufacturing industry was proposed to engage in emissions trading using blockchain technology in the [53] article. Public access to carbon emissions and reduction features would be available at every step of clothing production. Moreover, peer-to-peer trading frameworks are proposed in [54] to reduce carbon emissions in distribution networks and achieve regional energy balance. By analyzing prosumer bidding and selling prices, their algorithms can help determine the most efficient economic model for energy reshaping. In addition, authors proposed to segment the market based on reputation and order value priority in [55], [56]. They also conducted a case study and found that introducing reputation is more effective in achieving long-term carbon emission reduction.

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TABLE II
SUMMARY OF BLOCKCHAIN-BASED SOLUTIONS FOR CARBON EMISSION TRADING

Work	Major Contributions	Technical Approach	Focused Design	Limitations
[49]	A reputation-based ETS (Emission Trading System) system managed through distributed ledger. Two reputation-based mechanisms, i.e., market segmentation & priority-value-order.	Private permissioned blockchain, Multichain	Transparency, immutability, data accuracy, ETS efficiency and efficacy	Lacks detailed discussion on technical aspect with limited evaluation
[50]	A hybrid blockchain-based emissions trading scheme using Bancor AMM smart contract protocol, while maintaining a trusted authority role	Private or permissioned blockchain, smart contract	Transparency, tamper-resistance, security and high liquidity	General idea of tokens with no evaluation, lacks certificates trading details
[52]	A delegated PoS blockchain-based carbon emission rights verification system in P2P markets.	Public blockchain, smart contracts, Decentralized Application (DAPP)	Verification, anomaly detection, transparency	Design's technical details and evaluation are missing
[53]	Blockchain-based framework for assessing and measuring carbon emissions for the fashion industry.	Private/public blockchain, smart contracts	Transparency, automation, immutability, environmental/political performance	Application specific theoretical design, and no evaluation
[54]	An energy and carbon allowance trading framework is presented to achieve the energy balance of distribution networks.	Ethereum blockchain and smart contracts	Policy design, economical incentives, transparency, immutability	Focused more on energy trading. No technical details for carbon trading
[55], [56]	A reputation-based CET protocol where buyers are only able to view and interact with sellers that match their reputation criteria.	Public blockchain, smart contracts	Transparency, security, data immutability and trust	Significantly lack technical details and evaluation of the proposed design
[57]	A user-based carbon trading design to provide accurate carbon footprint from a multi-modal perspective, by recording them on a blockchain.	Public blockchain, smart contracts	Transparency, monitoring, verification	Privacy of vehicles' sensitive information is ignored. Limited evaluation
[58]	A blockchain-based method to trace the carbon footprint of products and organizations, to assist consumers in choosing products.	Public blockchain, smart contracts, DAPP	Tracing, monitoring, verification, transparency, trust	Lacks technical details and evaluation of the proposed design
[59]	Standardized firm's behaviors and transaction procedures to reduce carbon emissions through a trading model based on fluctuating reputation.	Public blockchain	Privacy, credibility, economic efficiency and security	Significantly lack technical details and evaluation of the proposed design
[60]	Proposed a Carbon Credit Ecosystem to digitize carbon credits in the form of NFT tokens for individuals and industries.	Public blockchain, smart contracts, DAPP	Transparency, accessibility, liquidity, and standardization	Significantly lack technical details and evaluation of the proposed design
[61]	A blockchain-enabled trading model based on delegated PoS consensus and smart contract, where bids and offers are characterized by the reputation value of the emitting enterprises.	Public blockchain, smart contracts, delegated proof-of-stake	Trading efficiency, security, transparency and trust	Focus more on blockchain consensus algorithm rather than actual emission trading
[62]	Proposed a science research based design to develop a blockchain-based architecture for emission trading scheme.	Ethereum blockchain, smart contracts, IPFS	Transparency, economic incentive, trust, immutability	Lacks evaluation of the proposed design
[63]	A qualitative analysis on blockchain-empowered platform to monetize, incentivise and realise the reduction of greenhouse gas emissions	N/A	Transparency, validation, monitoring, verification, economic incentive	A qualitative study rather than the technical contribution
[64]	Blockchain-based reputation mechanism to tokenize the carbon emission called Carboncoin tokens to trade with each other.	Public blockchain, smart contracts, ERC tokens	Transparency, monitoring, verification	Lacks technical detail and evaluation of the proposed model
[65]	A blockchain-based carbon trading system to enable automated auditing and the punishment of any violations that occur.	Private permissioned blockchain, smart contracts	Transparency, auditing, trust, monitoring	Lacks technical details and limited evaluation
[66]	A novel blockchain-enhanced trading system for construction industry to control carbon emissions within the project's materialization phase.	Public blockchain, smart contracts	Transparency, efficiency, decentralization, auditability	No system evaluation is provided for the proposed design
[67]	Proposed an emission trading system based on priority-based auction mechanism using permissioned blockchain and smart contracts.	Permissioned private blockchain, smart contracts	Transparency, monitoring, verification, economic incentive, trust	Lacks technical details and limited evaluation
[68]	Proposed a blockchain-based carbon trading and energy certificates market to increase participants' participation in load adjustment.	Private blockchain, smart contracts, consensus algorithm	Transparency, trust, economic incentive, security	Focused more on load adjustment, no details on emission trading process

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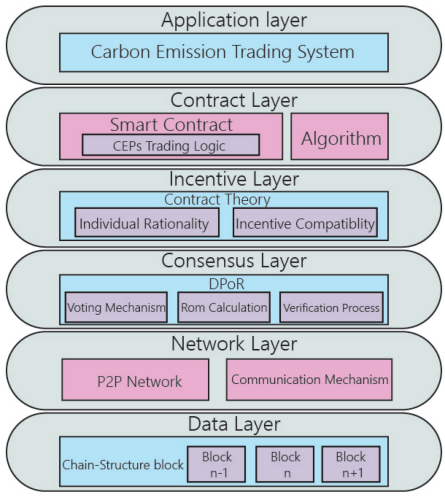


Fig. 5. BD-ETS framework proposed in [61].

In [57], the authors proposed a way to obtain accurate carbon footprints for individuals using multiple modes, which is necessary for the implementation of a user-based carbon trading scheme to reduce carbon emissions. Similarly, in [58], the authors argued that consumers can reduce carbon emissions by measuring their carbon footprints. Using a permissionless public blockchain, specifically Ethereum, they proposed a method for tracking and calculating carbon footprints. In [59], the authors proposed a Distributed Secondary Trading Model that uses blockchain technology to standardize firm behavior and transaction procedures, encouraging firms to take long-term measures to reduce carbon emissions. The model also includes a Fluctuating Reputation Value to inspire firms to invest in carbon emission reduction. Finally, in [60], in order to improve transparency, accessibility, liquidity, and standardization in carbon markets, the authors developed a Carbon Credit Ecosystem using smart contracts and blockchain technology. It includes an automated market maker for trading carbon tokens, as well as a mechanism for securely digitizing carbon credits.

In [61], the authors suggested a blockchain-based distributed system with a smart contract-based trading model and consensus mechanism to enhance the security and efficiency of carbon emissions trading. As illustrated in Fig 5, they proposed using a delegated proof-of-stake consensus mechanism, where the enterprise's reputation value depicts the power of voting to reduce the likelihood of an attack. In [62], the authors emphasized that fraud, lack of transparency, and insufficient economical benefits for emission reduction threaten the success of carbon trading systems. They suggested using the blockchain technology with the support of smart contracts to address these issues and ensure the system's success. Similarly, [63] proposed EnviroCoin, a platform that allows for carbon verification, validation, and monetization, which can serve as a business model to incentivize businesses and individuals to reduce carbon emissions. Additionally, A blockchain-based asset called CarbonCoin is proposed by the authors of [64], which allows energy producers to tokenize their carbon emission rights and exchange them for fiat currency. The authors also proposed using ESG data (environment, social, and governance) to create a reputation score that was more comprehensive. Finally, a motor vehicle restriction trading system called STRICTS based on blockchain is proposed in [65] based on the carbon emissions cap, which enables the automated auditing and punishment for emission violations without a trusted auditor.

In a recent study [66], a new blockchain-supported trading system was created for the construction industry to regulate carbon emissions during the materialization phase of a project. Smart contracts were utilized to manage the carbon emissions. Another study by [67] suggested a trusted and incentivized emissions trading system, using Hyperledger fabric, a permissioned blockchain, and smart contracts. The system provides a transparent trading of carbon credits with improved monitoring and verification of carbon credits. Finally, [68] proposed a blockchain-based market that incorporates carbon trading and RECs to encourage market participants to participate in load adjustment. The transactions are executed on a blockchain-based platform and combine power purchase transactions with green certificate transactions.

B. Blockchain-Based CET Platforms

Several blockchain-based currently available platforms to trade carbon credits are summarized below.

1) ClimateCoin: A blockchain-based platform called ClimateCoin encourages carbon reduction and sustainable practices. Through this program, individuals and organizations are able to earn tokens for reducing their carbon footprints. After the tokens are created, they can be exchanged or invested in renewable energy projects.

2) AirCarbon: AirCarbon is a blockchain-based marketplace for trading carbon credits. As a result, carbon offsets can be purchased and sold by businesses and individuals, making carbon trading more transparent and accessible.

3) Veridium and IBM: An easier and more effective way to reduce carbon dioxide emissions has been introduced by IBM and Veridium. Carbon offset credits are tokenized using blockchain technology by these companies. Veridium's sister company, InfinteEARTH, plays an important role in this process as well. A tokenized credit will be sold to different organizations that wish to reduce their carbon footprints.

4) ClimateTrade: ClimateTrade is a blockchain-based platform that facilitates the exchange of verified carbon offsets. It is a marketplace where buyers and sellers can trade carbon credits securely and transparently.

IV. Blockchain Contributions in Energy Certificates

In this section, recent works are examined to illustrate the use of blockchain in different kinds of energy certificates. A summary of blockchain-based energy certificates designs is provided in Table III.

A. Blockchain for Energy Certificates

The primary disadvantage of the current system for obtaining RECs is the lengthy and complicated application process, which

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TABLE III
SUMMARY OF BLOCKCHAIN-BASED SOLUTIONS FOR ENERGY CERTIFICATES

Work	Major Contributions	Technical Approach	Focused Application	Limitations
[69]	A hybrid blockchain-based REC model to trade REC certificates. A double auction mechanism to adjust the price of RECs and establish the trading rules	Consortium blockchain, smart meters, smart contracts	Renewable Energy Certificates	Lacks detailed discussion on technical aspect
[70]	A blockchain-based REC model to record RECs through smart meters directly. A subscription platform to trade RECs in exchange of tokens	Public blockchain and smart contracts	Renewable Energy Certificates	Lacks technical details and evaluation of the design
[71]	A trading mechanism based on Ethereum blockchain to simulate the marketplace using native tokens	Ethereum blockchain and smart contracts	Renewable Energy Certificates	Technical details are missing and system evaluation is limited
[72]	A blockchain-based mechanism to issue, receive and verify RECs without a trusted third party. A market-based approach is also used to regulate RECs' demand and supply	Ethereum public blockchain and smart contract	Renewable Energy Certificates	Provides a high level idea of REC issuance and verification without technical details
[73]	A blockchain-based proof-of-generation protocol with Igreencoin, as well as a trading system for RECs that can be used by prosumers and consumers.	ERC20 tokens, proof-of-generation consensus algorithm	Renewable Energy Certificates	No technical details regarding how certificates trading will happen
[74]	NRGcoin, a decentralized digital currency to trade locally produced renewable energy by the prosumers.	ERC20 tokens, Ethereum blockchain	Renewable Energy Certificates	Provides a high level overview of design and lacks technical details
[75]	A hybrid supply demand blockchain-based REC trading system, which utilizes double auction mechanism to improve the information transparency.	Permissioned blockchain, smart contracts	Renewable Energy Certificates	Blockchain integration details and system evaluation are missing
[76]	In the REC transaction market, seal-bid auctions are used to enhance security and maximize profit.	Public blockchain	Renewable Energy Certificates	Lacks information about how the personal information is secured, and how privacy is ensured
[77]	In the REC market, a mechanism is established to identify how different companies interpret and propose blockchain solutions to green energy problems.	Public blockchain	Renewable Energy Certificates	A qualitative study rather than the technical contribution
[11]	Providing small-scale energy producers with a trustworthy and anonymous REC trading mechanism.	Ring signatures, stealth address, trusted execution, public blockchain	Renewable energy certificates	Stealth addresses and Ring signatures are not implemented in this work
[78]	A blockchain-based architecture with a combination of ZKPs to provide transparency and privacy to green energy consumption and generation	Public blockchain, zero knowledge proofs	Renewable Energy Certificates	Only high-level ideas, missing system details and a security analysis
[79]	In order to ensure long-term security and transparency in energy certificate trading, blockchain technology was examined.	Ethereum blockchain, smart contracts	Energy white certificates, Energy performance certificates	Significantly lacks technical details and evaluation of the system
[80]	A permissioned blockchain system to facilitate reliable measurement and verification of energy performance, and automatic generation of EPCs.	Hyperledger Fabrics, chain codes	Energy performance certificates	A high level idea with limited technical details and evaluation
[81]	A blockchain-based trading platform to generate and issue RECs, by tokenizing RECs to issue, trade, verify and retire the certificates.	Permissioned blockchain, multi-chain, smart contracts	Renewable energy certificates	Lacks technical evaluation of the proposed model
[82]	From the perspective of cybersecurity, a technical evaluation of blockchain technology for tracking renewable energy.	N/A	Renewable energy certificates	A qualitative study rather than the technical contribution
[83]	Developed a blockchain-powered system for collecting and tracing core data, which conducts verification of traceability and on-chain certification.	Public/Private blockchain, smart contracts	Renewable energy certificates	Significantly lacks technical details and evaluation of the system
[84]	As part of Trinity project, this is a demonstration of Blockchain's potential for tracking REC lifetimes.	Stellar blockchain, Federated byzantine algorithm	Renewable energy certificates	Significantly lacks system's technical details and evaluation of the system

requires energy producers to register and prove their legitimacy. The application and approval process is also resource-intensive and not automated or intelligent. Additionally, the central authority has excessive power, including the ability to decide whether to issue certificates, oversee and record transactions, and disclose information, leaving room for abuse of power. The centralized nature of the system also makes it vulnerable to attacks, which can lead to the loss or tampering of information, resulting in unnecessary losses and disputes. The issuance of fraudulent certificates and double-counting is also a significant problem

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in the current system. Organizations like Greene attempts to address this issue through source audits, but these are costly for small-scale producers (an audit costs at least $4,500 per year). In addition, the current system is likely to violate privacy rights, especially for small scale traders and producers. In order to address these challenges, developing a traceable, secure, and trusted energy certification mechanism is critical. Several blockchain studies are summarized below to demonstrate how blockchain can be used to resolve the aforementioned issues.

Recently, Ming et al. [69] introduced a hybrid REC model based on consortium blockchain, which allows REC issuance, and REC trading using blockchain technology. Furthermore, the model suggests incorporating a double auction mechanism to regulate the price of RECs and set the rules for trading. The ARIMA model is also utilized to forecast the future prices of RECs. The issuance and trading of REC is done using smart contracts. Another major concerns with the article is the lack of detailed discussion on the technical aspects of the proposed solution, even though it attempts to address the issues in the current REC model. Shengan et al. [70] introduced blockchain technology into the current REC trading system to solve the potential challenges in the current design. All entities are connected to the blockchain network, and the smart contract generates the REC according to the power generated by the smart meter and recorded in the blockchain network. Moreover, they also propose a subscription platform, where the participants can list their green certificates and sell them to the buyers in exchange of the tokens native to the proposed platform. Although the paper provides a high level idea of blockchain-based REC trading system, it significantly lacks technical details and evaluation of the system.

A method for creating a simulated marketplace on the Ethereum Blockchain and Smart Contracts is presented by Castellanos et al. [71], that uses crypto-tokens to represent generated green energy/REC. The smart contracts within the REC marketplace enable prosumers to represent RECs as a token and allows them to sell it to other consumers. Cryptographic tokens that resemble RECs are traded by individuals without access to energy markets and a record of the balances of the participants is maintained by the smart contract. Two market strategies were simulated, i.e., fixed and variable price strategy, which allows tokens to be sold as RECs. For issuing, receiving, and verifying RECs, Knirsch et al. [72] proposed GECKO, a permissionless and decentralized system that integrates into existing infrastructures without relying on a central trusted third party. A market-based approach is utilized by this system to manage the supply and demand of green energy, and it enables end-users to authenticate certificates.

Zhao et al. [73] introduced I-Green, a system which is based on blockchain technology, to incentivize the adoption of renewable energy by offering individual green certificates to consumers and prosumers in the energy market. To facilitate voluntary green certificate trading, they introduced a green ratio scheme that calculates the renewable energy's percentage in whole energy consumption, both at the household and community levels. The proposed I-Green blockchain utilizes the Igreencoin as a native token and uses the proof-of-generation

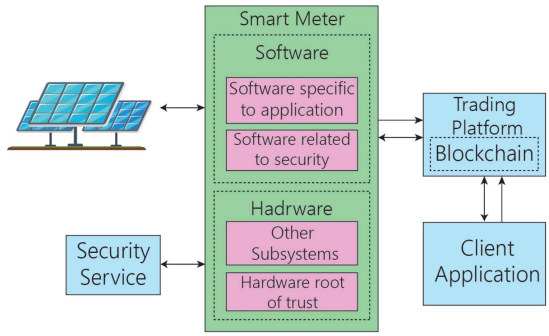


Fig. 6. System diagram proposed in [11].

consensus protocol. On the other hand, Mihail et al. [74] introduced NRGcoin, for trading locally produced renewable energy by prosumers. Under proposed design, renewable energy which is produced locally is converted into NRGcoins regardless of their value on the market. Agents send locally generated energy to DSO's lower-voltage substations via smart meters. However, the work provides only a high-level overview of the proposed design and lacks technical details.

Dong et al. [75] proposed a hybrid system for trading RECs that combines permissioned blockchain technology with the benefits of continuous double auction (CDA) and blockchain. The system uses blockchain technology in the issuance of RECs based on on-chain trading along with the off-chain bidding to facilitate REC data collection, review, and enhance issuance efficiency and transparency. Furthermore, the system incorporates CDA to facilitate REC transactions that are more market-oriented. This helps to increase transaction efficiency and allows for greater flexibility in adjusting the relationship between supply and demand, as well as customizing prices. A seal-bid auction technique utilizing blockchain technology was presented by Kim et al. [76] to maximize profits for participants in REC transactions while also addressing security concerns. As a result of their sealed bid second price auction, users' information is protected and buyers are able to identify honest bidding, while sellers have a reasonable profit margin. Despite that, the authors did not provide adequate information about how personal information of users is protected. Yamaguchi et al. [77] contended that the impact of blockchain technology on the REC market is not well understood, especially in terms of the effects on different organizations. They identified how differently-positioned organizations perceive their challenges and developed their blockchain technological artifacts for the REC market. The authors also analyzed the potential impact of blockchain on the Brazilian renewable energy market. However, the work mainly presents a qualitative analysis of the REC market from the perspective of different organizations and does not offer a technical contribution.

A new REC trading system developed by Dimcho et al. [11] addresses privacy and security concerns as depicted in Fig. 6. They propose a system that will provide small-scale energy producers with a secure and cost-effective way to participate in RECs. To achieve this, they utilize blockchain technology along with low-cost and secure hardware, stealth addresses, and

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ring signatures, to create RECs that are both anonymous and trustworthy. They proposed the use of ring signatures and stealth addresses to anonymize RECs, but they did not implement these in their work, leaving it as an open research problem. However, Johannes et al. [78] proposed a cryptographic zero-knowledge proof of RECs that is used to verify green energy generation RECs. Using their design, green energy labels for generation and consumption are more transparent and private. They provided transparency and verifiability by enabling on-chain verification of ZKPs using the stored Merkle roots. The proposed design does not reveal any information about the stakeholders involved, thus protecting privacy and data sovereignty. However, their paper only presented high-level ideas, and more detailed information about the system and security analysis of the solution are still missing.

Using blockchain technology, Asthma et al. [79] investigated how energy white certificates and energy performance certificates could be made more secure and transparent. Blockchain technology was used to address some of the key issues related to the UK Energy Company Obligation and white certificate scheme in Italy. In spite of this, the paper did not provide technical details or an analysis of the proposed system. A platform based on permissioned blockchain was also proposed by Ranathunga et al. [80] for managing energy in a district that considers the interaction between buildings in a district and their users. The platform enables trustworthy measurement and verification of energy performance and automates the DEC generation. The authors also provided details on the design and implementation of the platform. They proposed a three-tier architecture, evaluated their system in terms of various metrics, and the system's performance was measured using Hyperledger Caliper [85].

In a recent paper, Zuo et al. [81] proposed a decentralized blockchain-based platform to issue and trade RECs that provides improved traceability, transparency, and reduced costs for REC exchanges. The purpose is to represent RECs as blockchain tokens, allowing them to be issued, traded, verified, and retired trustfully. REC tokenization allows the platform to provide low-cost, transparent, and easy-to-use features. RECs are issued and tracked more reliably and securely with the immutable capture of trustworthy information on the blockchain. Additionally, Cali et al. [82] examine cybersecurity issues related to REC trading through the use of blockchain and distributed ledger technology from organizational perspective.

In [83], data collection and traceability systems were created utilizing blockchain's encryption and anti-tampering capabilities, which ensured the authenticity and credibility of the traceable data. Additionally, in [84], to issue RECs under the TRINITY project [86], authors proposed a blockchain-based platform. This platform is designed to demonstrate the potential of blockchain for tracking the lifetime of a REC, and this platform allows account holders, issuing bodies, and market operators to exchange data directly, securely, and automatically.

B. Blockchain-Based REC Platforms

Several blockchain-based currently available platforms to trade renewable energy certificates are summarized below.

1) Power Ledger: It is a well-known blockchain-based platform that facilitates peer-to-peer trade of renewable energy. Individuals and businesses can trade excess energy generated by renewable sources directly with each other.

2) WePower: WePower is a blockchain-based REC trading platform connecting energy buyers (households and investors) directly with green energy producers. By tokenizing energy, the platform provides a marketplace for trading energy tokens.

3) EnergiToken: Through EnergiToken, energy saving behaviour is recognized and rewarded. Tokens will reward energy-efficient behavior such as low-carbon transportation, buying energy-efficient appliances, and trading energy.

4) Grid+: With the aim of providing renewable electricity to consumers, Grid+ is building a blockchain-based energy trading platform. Grid+ plans to purchase wholesale energy and sell it directly to consumers who have created Grid+ accounts and pay in real-time through the Ethereum blockchain.

C. System Evaluation and Comparison

While there's been significant research on blockchain-based GHG emission reduction schemes, there's a lack of comprehensive literature that extensively evaluates these blockchain systems and compares them to traditional centralized GHG reduction approaches in terms of performance, security, scalability, etc. This review paper doesn't provide a quantitative analysis or comparisons between schemes. Instead, it offers explanations of the evaluation criteria for centralized and blockchain-based GHG emission reduction schemes, aiming to assist future researchers in conducting quantitative analyses and comparisons across various systems.

From a blockchain perspective, we need to assess transaction throughput, latency, scalability, and security. This includes evaluating the blockchain's capacity for processing trading transactions per second, confirmation times, scalability with participant numbers, and ensuring protection against fraudulent credits and transactions. Implementing GHG emission reduction systems on multiple blockchains and comparing their performance using these metrics can help fine-tune the blockchain for specific performance or security goals.

From a system perspective, we must compare blockchain-based solutions to traditional GHG emission reduction methods. Given the extensive participation in GHG emission trading, it's essential to evaluate and compare the efficiency of data processing, reporting, and verification between centralized and blockchain-based systems. Additionally, we should analyze the costs of maintaining centralized infrastructure and third-party audits to compare them with decentralized GHG emission reduction approaches. Lastly, quantifying the transparency levels in reporting and monitoring emissions data in both centralized and blockchain-based emission reduction schemes is necessary.

V. Current Research Challenges and Future Directions

This section briefly discusses possible research directions for the future, including how blockchain can be used for carbon trading and energy certificates.

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A. Security and Privacy

According to [87], privacy and security are significant issues in current energy systems based on blockchain technology. The information in the REC leaks privacy, especially for small-scale producers and traders who are unwilling to reveal their trades and level of market exposure. One of the open research challenges is to provide security and privacy to the traders in GHG emission reduction schemes. Simultaneously, transparent GHG emission reduction schemes are necessary to mitigate fraudulent trading activities, which could lead to the leakage of private information [11]. Therefore, another open research challenge is to find a trade-off between privacy and transparency, enabling the tracing of the origin of fraudulent activities or double spending in case of conflict. Privacy protection technologies, such as signature and encryption schemes, can address these security and privacy concerns. For example, [88] suggests that one approach to enhance anonymity in blockchain-based systems is by using a new blockchain address for each transaction. This method can make it more difficult to establish relationships among a user's transactions. Furthermore, there are additional techniques that can improve anonymity, including zero-knowledge proof [89], blind signature [90], ring signature [91], mixing service [92], and homomorphic commitments [93]. However, to ensure a secure and privacy-preserving blockchain-based trading and certification system while ensuring transparency is still an open research question.

B. Data Storage and Management

In scenarios involving carbon trading and energy certification, where the number of participating entities and transactions is significantly large, a substantial amount of data–both financial and non-financial transactions–needs to be stored in a tamper-proof manner and managed accurately. One requirement for this data storage and management is to openly share and verify emission and trading data, fostering trust among stakeholders. However, the challenge lies in achieving this objective while maintaining data privacy, which remains an open research question. Furthermore, it's worth noting that blockchains aren't inherently designed to handle large volumes of on-chain data [94]. This raises the research question of how to accurately and reliably store emission and trading data off-chain. To address this, decentralized off-chain data storage technologies like Filecoin, IPFS, Storj, and others can be employed. Nonetheless, storing energy and trading-related data directly on these solutions could potentially compromise privacy, given the paramount importance of data and identity privacy [95]. The reviewed research in this context doesn't offer comprehensive solutions to the data storage and management challenges within blockchain-based solutions. Therefore, the design of privacy-preserving and tamper-proof storage solutions for carbon trading and energy certification data requires further research.

C. System Scalability

With the increase in tokens, investors, users, and other entities, public blockchains are facing the issue of scalability, i.e., throughput, data size, response time, and cost, etc. [96], and these blockchain systems differ in terms of performance characteristics. In real-world energy applications, we have to deal with the large number of participating parties, which could lead to large number of transactions per second. Current blockchain systems are not designed to handle this large number of transactions and blockchain system scalability for energy applications is still an open research question. The scalability problem can be solved by using the private and consortium blockchain with customized consensus mechanisms [97], however, it comes at the cost of decentralization offered by the public blockchain. The scalability problem can also be solved by using layer-2 blockchain solutions, which implement off-chain computation techniques to increase the overall scalability of the system [98]. The research work covered in this review significantly lacks the implementation of the proposed designs and does not analyze the scalability aspect of the proposed designs. Therefore, designing, implementing and analyzing the blockchain-based carbon trading and energy certification mechanisms is still an open research question.

D. Decentralization and Legal Governance

The management of carbon trading and energy certificates is typically overseen by regulatory bodies who act as intermediaries between the government and stakeholders such as businesses, citizens, communities, and non-profits [99]. When implementing a blockchain-based system, it's important to include these regulatory bodies in the design process and ensure appropriate authorization to prevent violations of rights and privileges. Given that these systems often begin with a centralized approach, transitioning them to a decentralized model demands a formal analysis of design requirements to comprehend the fundamental needs of blockchain-based systems, such as desired levels of decentralization and privacy considerations. Additionally, auditability is an important consideration to reconstruct events or actions from historical records for liability purposes, regulatory compliance, or financial interests. However, since blockchain is inherently decentralized, designing a framework that incorporates regulatory bodies can be a challenge as it goes against the principle of decentralization. Therefore, it is imperative to pinpoint the necessary design criteria for blockchain applications, e.g., transparency, traceability, security, privacy, and others. These criteria should also be capable of integrating regulatory bodies into the frameworks governing carbon trading and energy certificates. This domain necessitates additional research and consideration moving forward.

E. Standardization

The blockchain-based carbon trading and energy certification systems are being developed with various mechanisms, protocols, and technological solutions. However, the lack of commonly accepted standards presents a major challenge in the integration of IoT devices, smart meters, and other related entities. As these certificates are traded among countries, a decentralized emission and energy trading standard is still an open research question. In the absence of a trusted intermediary or centralized authority, standardization efforts are crucial to mitigate the conflicts among trading parties and to ensure trust

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among stakeholders, industries and countries. Therefore, further research is needed to achieve blockchain standardization in areas such as communication exchange, advanced security and privacy, reward/penalty policies, etc. to ensure smooth interoperability.

VI. Conclusion

Currently, many countries around the world plan to significantly reduce greenhouse gas emissions by promoting different emission reduction systems. However, these proposed systems are centralized, highly regulated, and operationally expensive, and do not meet the transparency, trust and security requirements of the system. Blockchain is an ideal solution for such problems due to its decentralized, transparent, trustworthy, and anonymous nature. IThis article provides a comprehensive review of how blockchain has been used in reducing GHG emissions through various schemes such as CET, RECs, and TWCs. Finally, the article also highlights recent contributions of blockchain in GHG emission reduction systems and outlines some of the challenges that need to be addressed in future research. We hope that this article will open up several opportunities for further research in this field.

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